Earth and Planetary Science Letters 375 (2013) 123–134

Contents lists available at ScienceDirect

Earth and Planetary Science Letters

journal homepage: www.elsevier.com/locate/epsl

Upper Permian magnetic stratigraphy of the lower Beaufort Group, Karoo Basin

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ARTICLE INFO

Article history:
Received 29 August 2012
Received in revised form
25 April 2013
Accepted 9 May 2013
Editor: J. Lynch-Stieglitz
Available online 20 June 2013

Keywords:
magnetic stratigraphy
geochronology
Late Permian
Karoo Basin
Illawaran interval
Kiaman superchron

ABSTRACT

We carried out a magnetostratigraphic and geochronological study of late Permian sediments in the Karoo Basin of the Western Cape Province, South Africa. A continuous, ~700 m thick section of deltaic sediments of the upper Waterford Formation (uppermost Ecca Group) and the fluvial sediments of the Abrahamskraal Formation (lowermost Beaufort Group) were sampled at the meter scale. U–Pb dating of zircons from interbedded volcanic ash beds by ion microprobe (SHRIMP) provided absolute age constraints on the age of the sedimentary rocks. Paleomagnetic analysis reveals a partial overprint of the Natural Remanent Magnetization (NRM) that is tentatively ascribed to the emplacement of the Karoo Large Igneous Province in the Western Cape region during the middle Jurassic. A stable component of the NRM was found at temperatures higher than 450 °C and was interpreted as a Characteristic Remanent Magnetization (ChRM) acquired during deposition, supported by a positive reversals test for this dual polarity ChRM. The virtual geomagnetic pole position for the Waterford and Abrahamskraal Formations computed from the average ChRM direction is in general agreement with the late Permian directions for stable Gondwana. A significantly different average inclination, and thus paleomagnetic pole position, is obtained by correcting the inclination shallowing error by the Elongation–Inclination method (Tauxe and Kent, 2004). The presence of both normal and reversed polarity zones indicate deposition after the end of the Kiaman Superchron, moreover the polarity sequence is in good agreement with the Illawarra sequence of Steiner (2006). Our results indicate a Capitanian (late Guadalupian) age for the Abrahamskraal Fm., in agreement with the Late Permian age, based on presence of *Glossopteris* flora and *Dicynodont* fauna, traditionally assigned to the fluvial-lacustrine sediments of the Beaufort Group. However, the U–Pb zircon ages of ca. 264–268 Ma suggest an age of 269 Ma for the top of the Kiaman superchron.

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1. Introduction

The Kiaman superchron, also known as the Permo-Carboniferous Reverse Superchron, marks a long, 40–50 Ma period in which the geomagnetic field was stuck in a reversed polarity, first noted by Irving and Parry (1963) in samples near the town of Kiama in New South Wales, Australia. Opdyke et al. (2001) established the base of the Kiaman interval as 319 Ma. The return to a geodynamo with aperiodic but regular reversals was observed in late Permian sedimentary rocks from the Illwara region in the Sydney Basin of eastern Australia, so this post-Kiaman series of short, normal and reverse magnetozones is known as the Illawaran series. The age of this series is not well-established, generally due to the lack of integration between geochronological and palemagnetic studies of sedimentary rocks. At present, the best estimate for the age of the base of the Illawaran series is ca. 265 Ma (Glenister et al., 1999), based on zircon age data reported in Bowring et al. (1998) from the Guadalupe Mts. of west Texas.

Unlike the highly precise chronology of the latest Permian (Lopingian) to early Triassic, which has been established from integration of biostratigraphy and U–Pb geochronology (Bowring et al., 1998; Mundil et al., 2004; Shen et al., 2011), as well as carbon isotope chemostratigraphy (e.g., Korte and Kozur, 2010) from multiple sections worldwide, magnetostratigraphy provides no such consensus for the preceding late Middle Permian (Guadalupian) times. Like the end-Permian mass extinction, the end-Guadalupian epoch is also marked by a dramatic decrease in biodiversity (Stanley and Yang, 1994; Sahney and Benton, 2008), so establishing a global chronological reference frame is important for correlation purposes. The Karoo Basin is the site of one of the

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http://dx.doi.org/10.1016/j.epsl.2013.05.017

thickest and best-exposed records of sedimentary deposition through late Paleozoic and early Mesozoic times, and it is abundantly endowed with the vertebrate fauna that define the continental record of the late Permian and early Triassic (Smith and Ward, 2001; Rubidge et al., 2013). Our study will address the viability of magnetostratigraphy in the Karoo Basin, as well as establish the Guadalupian age of the Beaufort Group, in contrast with the Permo-Triassic age suggested recently by Fildani et al. (2009).

1.1. Geological setting

The Karoo Basin of South Africa records a long history of sedimentation throughout the Paleozoic to middle Mesozoic with the accumulated deposits reaching up to 5500 m in thickness (Tankard et al., 2009). The Karoo Basin formed on African continental lithosphere, 1000–1500 km removed from the Panthalassan (i.e. palaeo-Pacific) plate margin, with both deposition and erosional intervals that can be broadly correlated with the other Gondwanan cratons. By late Cambrian times, following the stabilization of southern Gondwana, sedimentation along the Panthalassan margin initiated with deposition of the Table Mountain Group, a thick package of Cambro-Ordovician quartz arenites, that are overlain by Devonian-Carboniferous rocks of the Witteberg Group. The Permo-Carboniferous glaciation of southern Gondwana is recorded by the Dwyka Group diamictites, which is superseded by Ecca Group slope and delta deposits.

Following the deposition of the Ecca Group turbidites and deltaic sandstones and mudstones, a transition to a continental setting is inferred from the fluvial sediments of the Adelaide Subgroup of the lowermost Beaufort Group. These rocks are typical of a meandering river systems, with fine grained mud- and siltstones beds, 1–5 m thick, interbedded with coarser grained, 5–10 m thick channel fill fine-grained sandstones. The upper sedimentary units of the Beaufort Group are known as the Tarkastad Subgroup, which includes predominantly thick sandstones (>10 m beds) of the Katberg Fm. These rocks are considered to mark a transition to faster-flowing rivers with braided geometries, possibly linked to either faster erosion of a denuded continental surface (Smith and Ward, 2001) or runoff from the tectonically-uplifted Cape Fold Belt. During the Jurassic, continental flood basalts with concomitant dyke and sill intrusions (ca. 180 Ma, Duncan et al., 1997; Jourdan et al., 2007) and extensional tectonics resulted in the termination of Karoo Basin sedimentation (Tankard et al., 2009).

1.2. Previous results

The late Paleozoic and early Mesozoic sedimentary sequences of the Karoo Basin represent an attractive target for paleomagnetic studies. Many early paleomagnetic studies of the Beaufort Group report pervasive remagnetization, which was generally ascribed to the thermal effects of the emplacement of the Karoo Large Igneous Province during the Jurassic. For example, Ballard et al. (1986) carried out a study of the Lower Beaufort Group using standard paleomagnetic techniques that included chemical demagnetization. Their samples were thermally demagnetized up to a temperature of 450 °C and alternating-field demagnetized at a maximum field of 90 mT, showing single component NRM with an average direction 340°E and inclination −58°, which was interpreted as secondary magnetization after a negative fold test. Unreliable paleomagnetic results were also reported from unpublished studies in the Upper Beaufort Group red beds exposed in the eastern part of the Karoo Basin (Graham, 1961; Graham and

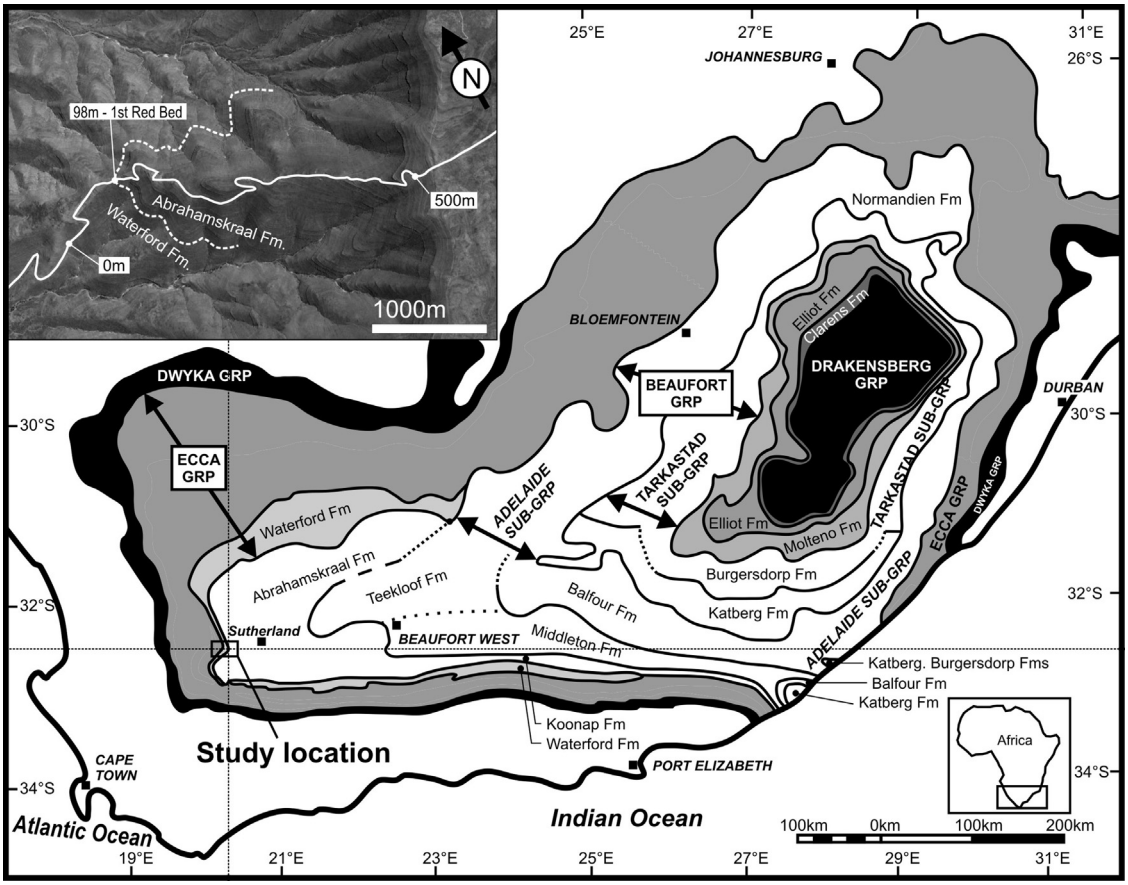


Fig. 1. Map of the studied area. Geological map after Johnson et al. (1997).

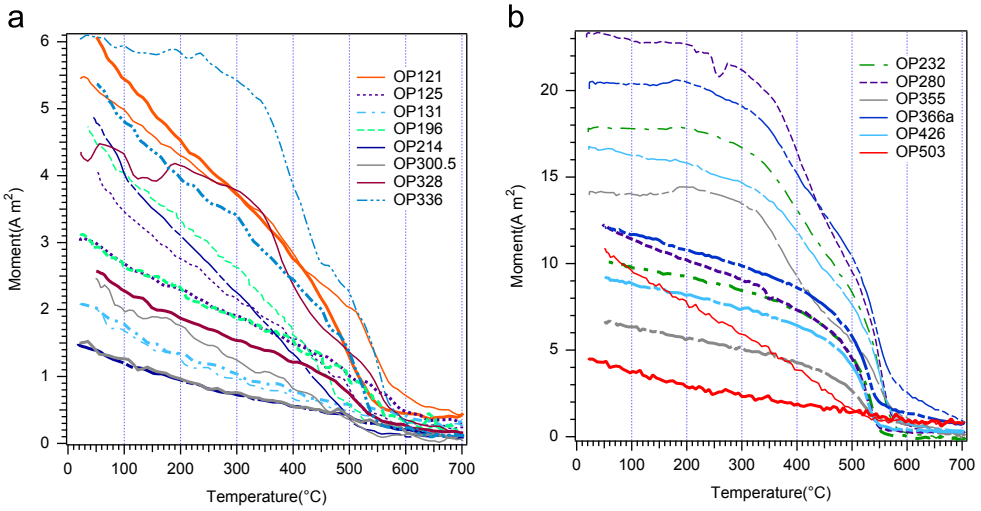


Fig. 2. Thermomagnetic curves of typical samples from Abrahamskraal Fm. Thick and thin lines indicate warming and cooling cycles respectively. Thermo-remanent curves measured with the VFTB indicate Curie temperatures between 550 °C and 600 °C in all measured samples suggesting that magnetite is the main magnetic carrier in the samples from Ouberg Pass. The concave curve shown by several specimens indicate the presence of large amount of paramagnetic minerals, which is confirmed by high-field measurements. Several samples who underwent thermal alteration during the measurement gave different heating and cooling curves, with the latter showing a higher magnetization. Given the large amount paramagnetic minerals it is conceivable to assume that part of the iron from paramagnetic minerals is oxidized to magnetite/maghemite at high temperatures. The two separate graphs are used for samples with (a) low and (b) high magnetic moment.

Hales, 1961), as well as Early Triassic Maji ya Chumvi sandstones in Kenya (McElhinny and Brock, 1975), discouraging further investigations. Only a few reports in the literature recognized that the Western Cape region has suffered only a partial remagnetization. Primary magnetizations for the Karoo basin sediments were reported in the study by Bachtadse et al. (1987), which yielded primary paleomagnetic poles from Paleozoic formations in the western Cape Fold Belt in order to improve the apparent polar wander path of Gondwana. A paleomagnetic pole from the Permo-Carboniferous Dwyka Fm. was also reported by Opdyke et al. (2001). Magnetic stratigraphy of the Permo-Triassic sediments in the central Karoo basin was reported by De Kock and Kirschvink (2004) and by Ward et al. (2005).

Age estimation of the Ecca Group and Beaufort Group are mainly based on pollen, flora and vertebrate assemblages, the latter believed to be one of the most complete terrestrial successions up to the Permian–Triassic boundary (e.g., Smith and Ward, 2001; Retallack et al., 2006). Radiogenic isotope dating of stratigraphic units in the Beaufort Group is, therefore, of utmost importance for time-stratigraphic correlation with other successions worldwide, but it is unfortunately poorly constrained. As mentioned before, Fildani et al. (2007, 2009) obtained U–Pb SHRIMP zircon ages of ~255 Ma for the upper middle Ecca Group (Skoorsteenberg and Laingsburg Formations), suggesting a much younger age than previously estimated for the overlying Beaufort Group and a potentially lower stratigraphic position of the Permo-Triassic boundary.

2. Paleomagnetism of the lower Beaufort Group

2.1. Lithology and sampling

The studied section is located at Ouberg Pass (GPS coordinates S 32° 24.44′, E 20° 19.58′) along the gravel road that ascends the Great Escarpment (Fig. 1). The section encompasses the upper part of the arenaceous Waterford Formation (deltaic sediments) at the top of the Ecca Group and the lower 400 m of the Abrahamskraal Formation (fluvial sediments) of the Lower Beaufort Group (Adelaide Subgroup), a total thickness of 700 m. Rossouw and De Villiers (1952) defined the Abrahamskraal/Waterford contact as the first appearance of red/purple mudstones in the succession. Whilst this does not necessarily represent the transition from delta top to alluvial plain (Johnson, 1966; Jordaan, 1981) it was considered to represent the first subaerial deposition. We follow the more practical criteria of Rossouw and De Villiers (1952) in defining the base of the Abrahamskraal Fm. (98 m mark in our stratigraphic profile) as the first appearance of red mudstones recorded in our stratigraphic logging. The Abrahamskraal Fm. consists of mudstones, including incipient paleosols and subordinate, very fine- to medium-grained, arkosic sandstones.

Macroscopic evidence of pervasive tectonic strain (e.g., folding or cleavage) from the Cape Belt 100 km south is rare or absent at the study section. The regional bedding dips $>5^{\circ}$ toward the northeast (320/02 according to measurements from Laser Imaging Detection and Ranging models acquired of the study section), and can be considered practically horizontal. A ca. 40 m thick thrust-fault zone was found at section depth between −120 m and −160 m, i.e., below the contact between the Waterford and Abrahamskraal Fms. This deformed zone included a 2 m thick highly deformed shear zone with an imbricated duplex structure along the fault plane (strike 340° dip 15°). Reverse displacement has resulted in 10 m scale fault-related folding, with asymmetric drag folds in the duplexed fault core that recorded top-side up motion. Deformation affected the paleomagnetic directions from samples within the fault zone, which appear remagnetized along anomalous directions and are discarded from further consideration or discussion.

Paleomagnetic samples consist of standard 2.5 cm diameter cores, which were taken from all outcropping lithologies except the most friable paleosols at an average interval of ca. 1 m using a gasoline-powered drill and were oriented using magnetic and solar compass.

Bentonitic ash beds were collected from well-defined beds throughout the Abrahamskraal Fm. The bentonite beds are 5–15 cm thick and found only in contact only with mudstones. The ash beds are characteristically white to light-gray, and clay-rich, providing a clear contrast to the green or red, silty mudstones of the Abrahamskraal Fm. There is no evidence for reworking of the bentonite layers in the mudstone successions by the low

energy processes inferred to form the thick mudstone layers, which are interpreted as overbank deposits. These horizons were sampled so as to avoid contamination by surrounding mudstones, and heavy mineral separates were prepared by standard heavy liquid techniques at the University of Stellenbosch.

2.2. Rock magnetism

The magnetic mineralogy was studied by unblocking temperatures and coercivity. The variability of magnetic properties along the section was investigated by measuring the magnetic susceptibility in the field with a SM-20 portable magnetic susceptibility meter as well as saturation IRM at the field of 2 T and ratio between IRM acquired at 0.3 T and 2 T (S-ratio) on a subset of selected samples. All remanent magnetization measurements were performed using a 2G DC-SQUIDs magnetometer. IRM was induced in the specimens using a ASC Pulse Magnetizer and NRM demagnetization was performed using 2 G alternating field demagnetizer, a ASC TD48 thermal demagnetizer (at the Alpine Laboratory of Paleomagnetism) and a Magnetic Measurements TD80A thermal demagnetizer (at the University of Western Australia). Hysteresis loops and thermomagnetic curves for Curie temperature investigation were measured using the Petersen Instrument, Variable Field Translation Balance at the paleomagnetic laboratory of the University of Western Australia.

A selection of thermomagnetic curves is shown in Fig. 2. Nearly all samples underwent some alteration after heating to the maximum temperature of 700 °C as suggested by the cooling curves that rarely overlap with the heating curve. However, a Curie temperature around 550 °C is clearly identifiable in most of the samples, which do not exceed the Curie temperature for stoichiometric magnetite (580 °C). Despite the relatively wide range of lithologies, the Curie temperature shows little variability along the section without any significant trend, indicating a constant magnetic mineralogy. A strong paramagnetic behavior is observed in the majority of the samples, characterized by a low magnetization intensity (Fig. 2a). This is corroborated by hysteresis loop measurements, which have generally shown a very large, high-field susceptibility suggesting that the low-field susceptibility is dominated by paramagnetic minerals in these samples. On the other hand, a substantial ferrimagnetic contribution to low-field susceptibility seems very likely in other samples because of their very high susceptibility and IRM values.

The large variability in the concentration of magnetic minerals along the section is shown by the susceptibility profile (Fig. 3). The low and constant magnetic susceptibility of the upper Waterford and lower Abrahamskraal Formations contrasts with the sharp peaks and the mean higher values of the upper Abrahamskraal Fm. This general behavior is backed by a similar pattern observed in the IRM profile which points to an increased supply of ferrimagnetic mineral starting at the section depth of about +160 m.

The S-ratio profile confirms the dominant presence of low-coercivity minerals with only a few samples dominated by a higher coercivity phase, mostly occurring in the upper part of the section.

The tensor of Anisotropy of Magnetic Susceptibility (AMS) was measured in a subset of samples from the Abrahamskraal Formation using a KLY-3 susceptibility meter and the standard, 15-positions orientation protocol. Samples generally have a weak anisotropy and the shape of the AMS ellipsoid is mostly oblate with maximum and intermediate axes lying close to the bedding plane, as often found in sedimentary rocks (Fig. 4a). Despite the small degree of anisotropy, the pattern of principal axes of magnetic susceptibility shows a well-grouped N–S direction of the maximum anisotropy axes which are imbricated toward the north (Fig. 4b). The mostly oblate, weak magnetic fabric measured

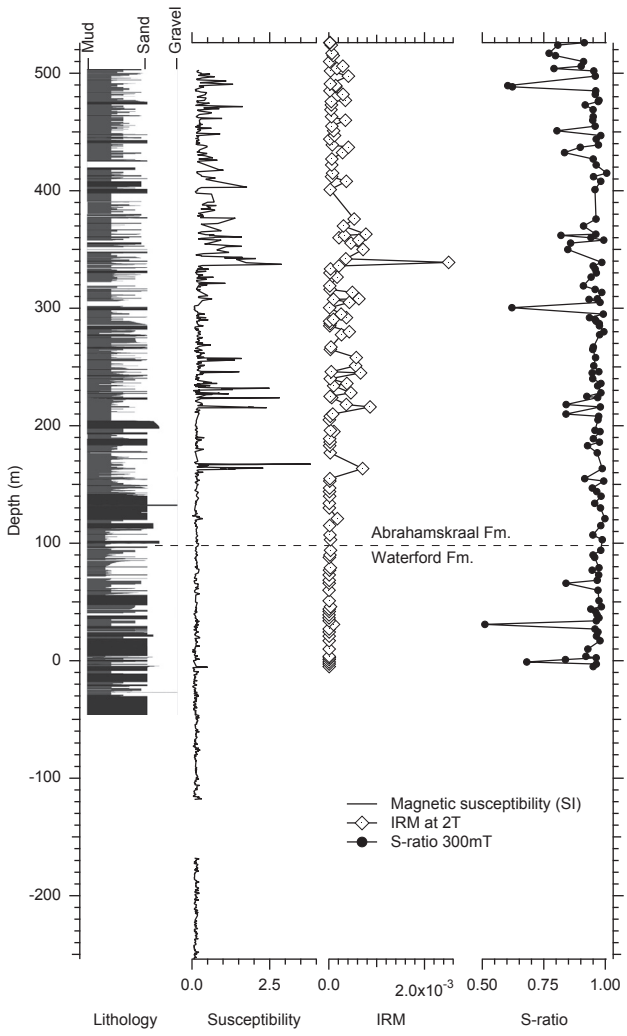


Fig. 3. Lithology, magnetic susceptibility, Isothermal Remanent Magnetization and S-ratio profiles of the Ouberg Pass section. The detailed susceptibility profile was measured in the field, the IRM and S-ratio were measured in a subset of samples from the Abrahamskraal Fm.

by AMS shows minimum susceptibility axes (K-min) dispersed around the vertical direction as expected for sedimentary fabrics. However, the close clustering of the maximum susceptibility axes (K-max), indicated by the area contour in Fig. 4b, suggests some process at work other than pure compaction.

The positive reversal test indicates that the effect of this weak preferential orientation of the grains on the ChRM directions is negligible. Considering the prevalent E–W trend of the tectonic structures in the closest section of the Cape Fold Belt we are tempted to suggest that the well grouped K-max, imbricated toward the North, represents an indicator of paleocurrents (e.g., Hrouda, 1982). Similar paleocurrent directions to our AMS fabric direction were reported for the Karoo region east of longitude 24°E (Cole and Wipplinger, 2001; Theron, 1973). However it is difficult to reconcile paleocurrent directions we measured from sedimentary structures within sand bodies in the investigated area, showing a mean flow direction north-east, to paleocurrent measurements determined from the AMS fabric. Although scatter in paleocurrent from the mean regional slope direction is normal for meandering stream sedimentation, the ca. 40° difference between the mean direction inferred from sedimentary structures and AMS fabric seems significant and cannot be explained with reference to fluvial processes. The meaning of the AMS pattern remains thus uncertain.

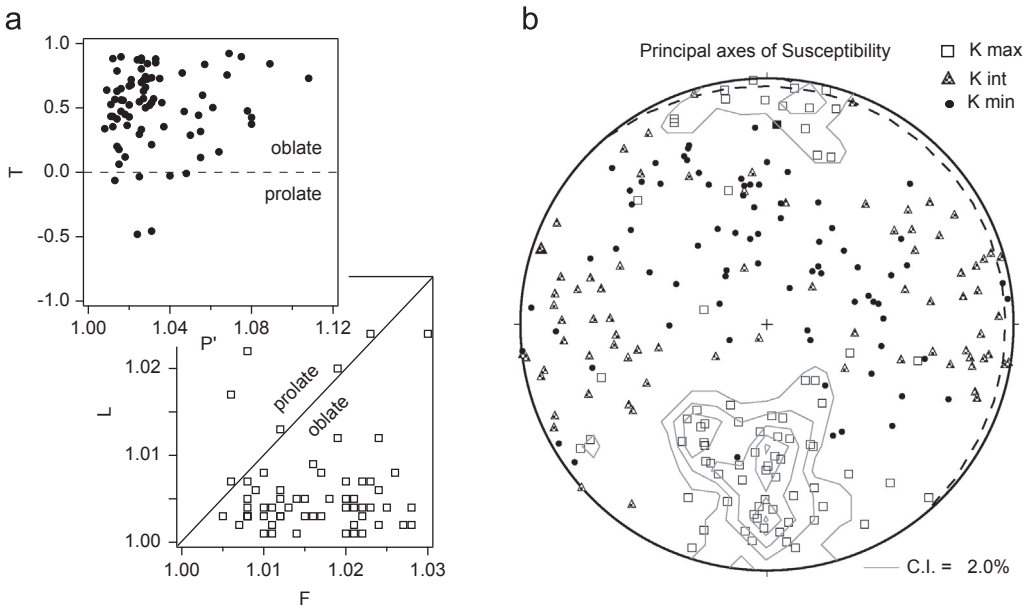


Fig. 4. Pattern of principal axes of magnetic susceptibility in the Abrahamskraal Fm. (a) Jelinek and Flinn diagram indicating a generally oblate shape of AMS ellipsoids, which is possibly a consequence of sediment compaction. (b) Equal-area plot (lower hemisphere) of the tilt corrected directions of the principal axis of AMS, the dashed great-circle represent the regional bedding plane. Despite the oblate ellipsoids the major AMS axes are well grouped in the North–South direction and imbricated toward the north, as highlighted by the 2% area contour lines of K-max (gray lines).

2.3. Characteristic remanent magnetization (ChRM) and polarity sequence

Natural Remanent Magnetization (NRM) was studied by progressive demagnetization. Alternating field and thermal demagnetizations were both tested in a set of pilot samples from the Abrahamskraal Fm. Alternating field demagnetization was generally unable to fully demagnetize the NRM and to isolate its most stable component. The best results were obtained by thermal demagnetization, and so this was used as standard procedure. Paleomagnetic directions were determined by least-squared line-fitting using principal component analysis, avoiding the use of great circles.

The NRM comprises two distinguishable components that are interpreted as ancient. A present day field overprint is commonly absent or completely removed by 150 °C. A more consistently observed paleomagnetic component is removed by stepwise heating to temperatures <450 °C and AF peak field below 90 mT, and is hereafter referred as component-A. Component-A is well grouped and has always a normal polarity with a Fisher mean declination 348°, and inclination −61° (Fig. 5a). The mean component-A direction is indistinguishable from that reported by Ballard et al. (1986) for the Lower Beaufort Group (see Table 1), which was isolated within an identical range of temperatures and alternating fields.

A second stable magnetization component was isolated by thermal demagnetization at temperatures ranging from 450 °C to 580 °C and is referred to as Component-B. A small number of samples, probably hematite-bearing, show maximum unblocking temperatures for component-B higher than 600 °C (Fig. 6a). Not all samples exhibit both components A and B even using the more effective thermal demagnetization; component-B is less common than lower temperature component-A, being present in $<40\%$ of the measured samples. Most discarded specimens were either completely remagnetized or did not show a stable (strongly aligned) direction in the unblocking temperature range expected for the component-B (Fig. 6b). Another relatively large group (ca. 20) of samples showed an extremely well-aligned single-component NRM along the whole range of temperatures associated with generally high NRM intensity, random declinations and very shallow inclinations (Fig. 6c). These samples were interpreted as possibly remagnetized by lightning and were discarded from further consideration.

The component-A of the NRM shows only normal polarities (mean direction 348/−61, Table 1), in contrast, the B component of the NRM displays both normal and reversed polarities (Fig. 5b) with antipodal directions that pass both Watson's and Monte-Carlo reversal tests (Fig. S1). The reversed polarity predominates in the B-component, and can be clearly distinguished from the component-A in the analysis of the vector plots. However, the similar directions of the normal-polarity component-B with the component-A led us to distinguish the putative normal-polarity component-B based on its unblocking temperatures greater than 450 °C.

The component-B was interpreted as Characteristic Remanent Magnetization (ChRM). The mean paleopole directions of the ChRM and of the A-component are compared with literature data in Fig. 5c and the resultant magnetic polarity sequence is shown in Fig. 7. Magnetic polarity stratigraphy of the Ouberg Pass section has identified 3 normal and 4 reversal magnetozones, with a magnetozone defined by two or more stratigraphically consecutive samples with the same polarity as indicated by the latitude of the virtual geomagnetic pole (VGP) (Fig. 7). Magnetozones defined by a single sample were considered uncertain. The short normal-polarity zone at the bottom of the section at about −190 m depth was interpreted as a repetition of magnetozone N3 due to the effect of the above thrust-fault. The inferred stratigraphic offset of $\sim$190 m would correspond to a throw of $\sim$600 m for this fault zone.

The paleomagnetic pole position computed from the average ChRM direction is in agreement with the paleopole position from the Permian/Triassic section of the continental Karoo (De Kock and Kirschvink, 2004). However, we have used the Elongation–Inclination method of Tauxe and Kent (2004) to correct for potential inclination shallowing error (Fig. S2) and found a relatively large ca. 9° shallowing error in the average inclination, which result in a significantly different paleopole position as listed in Table 1 and shown in Fig. 5c.

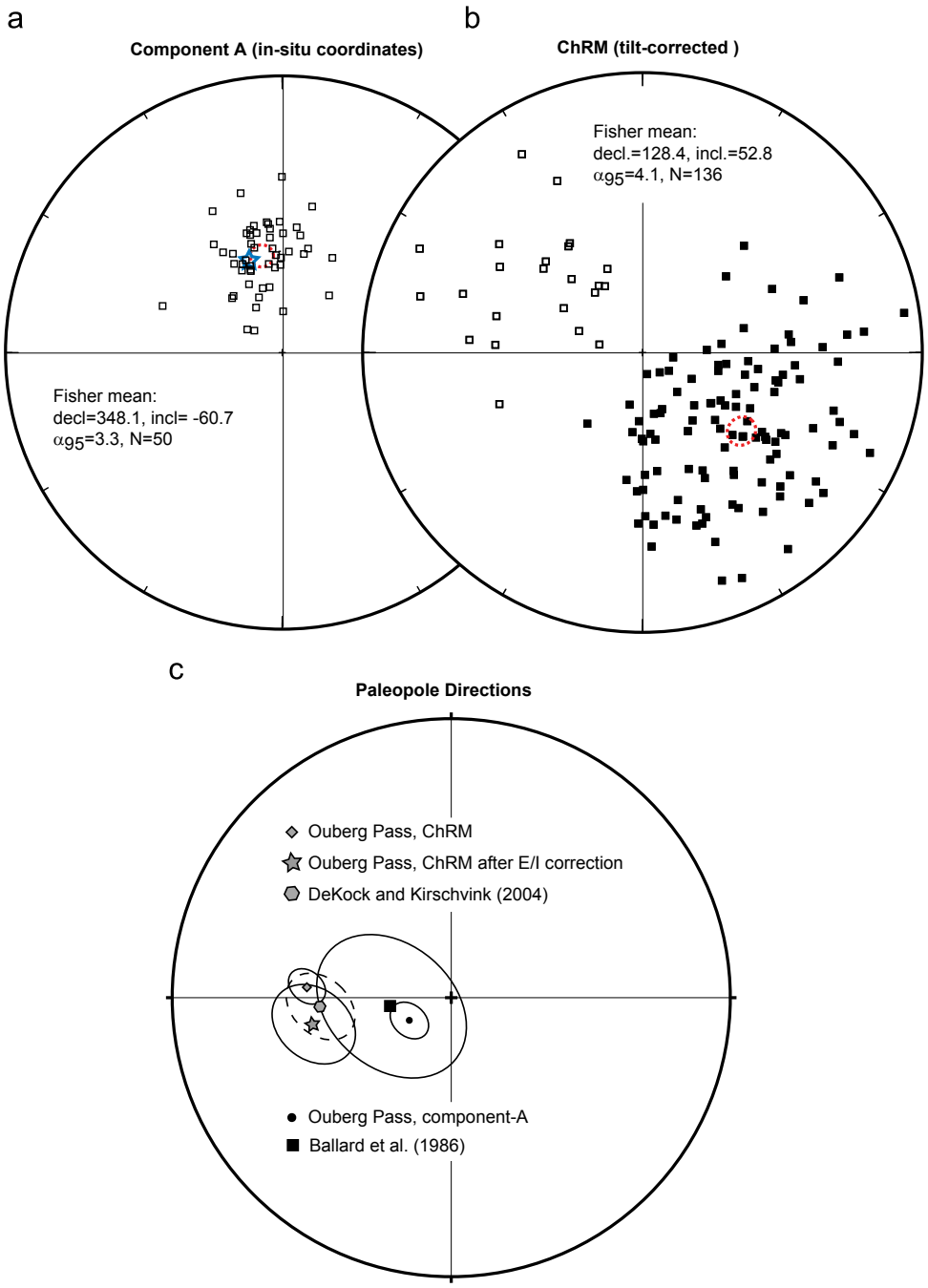


Fig. 5. (a) In situ directions of the A-component, isolated at unblocking temperature below 450 °C, from the Abrahamskraal Fm. at the Ouberg Pass section. The red-dashed circle represents the 95% confidence area of Fisher mean direction and the blue star is the mean direction of Ballard et al. (1986). (b) Tilt-corrected direction of ChRM from the Ouberg Pass section. The red-dashed circle represents the 95% confidence area of Fisher mean direction. Normal and reversed directions are antipodal and pass a reversal test. (c) Paleopole directions and 95% confidence level of the ChRM and A-components compared with literature data. The confidence level for paleopole of De Kock and Kirschvink (2004) (dashed) was not reported in the literature and was calculated from published data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Mean directions (Fisher) for Ouberg pass section compared to published data. VGP position is computed from the Fisher-mean paleomagnetic direction.

	Dec.	Incl.	k	α_{95}	N	VGP Lat.	VGP Long.
A-component	348	−61	38.6	3.3	50	N 76	E 241
Ballard et al. (1986)	340	−59	16	16	37	N 72	E 262
B-component Normal	303	−51	12.4	9.0	23	–	–
B-component Reversed	130	53	9.2	4.6	113	–	–
B-component Combined	308	−53	9.6	4.1	136	N 47	E 274
E/I Corrected (Tauxe and Kent, 2004)	308	−62	–	8.5	136	N 48	E 260
De Kock and Kirschvink (2004)	312	−62	17.8	7.6	35	N 51	E 266

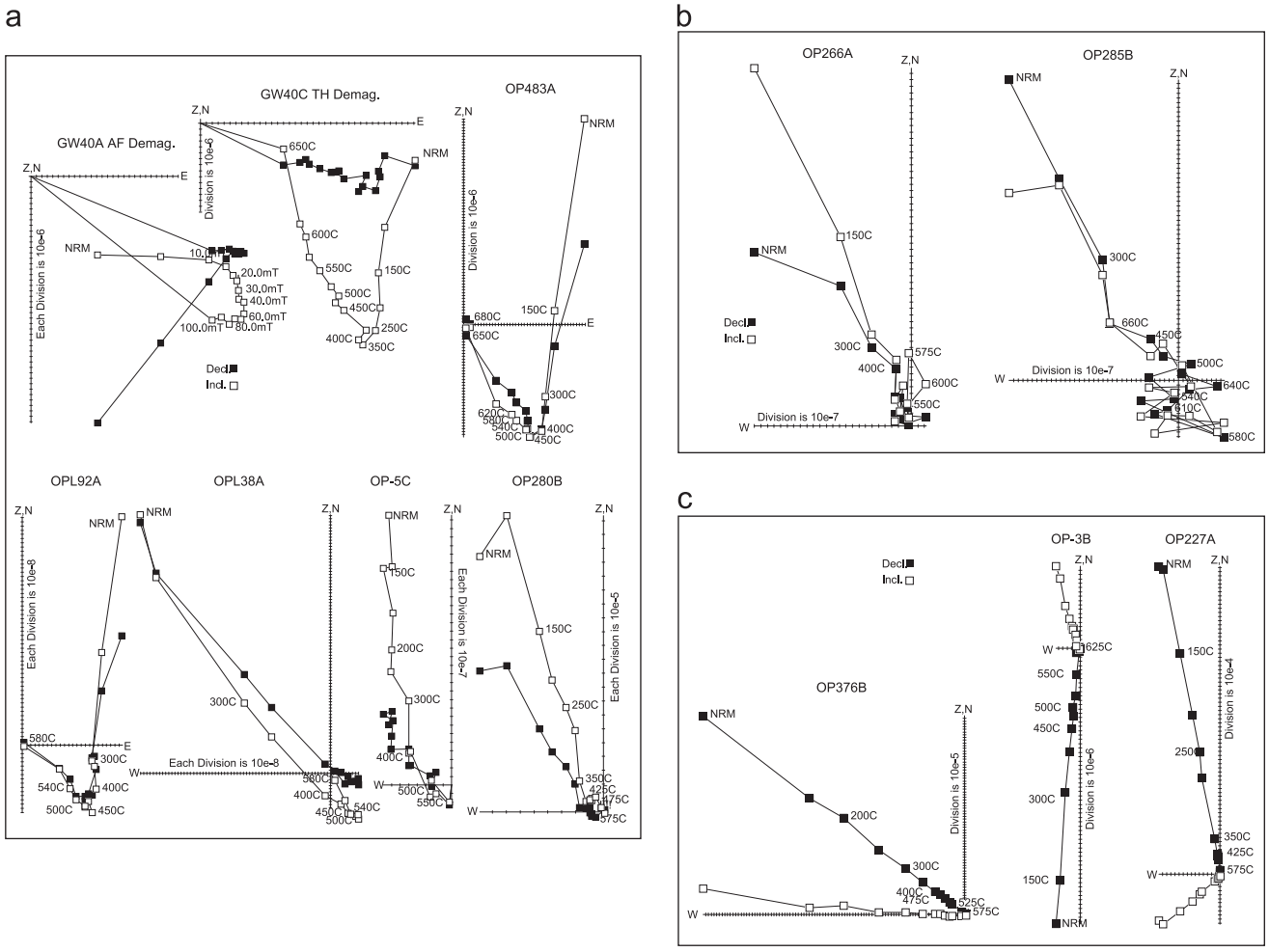


Fig. 6. (a) Representative examples of orthogonal vector plots form the Abrahamskraal Fm. Components A and B can be clearly identified in most of the demagnetization diagrams. (b) Specimens completely remagnetized or that do not show any stable direction above 450 °C. (c) Instance of specimens with single-component NRM and very shallow inclinations that were interpreted as remagnetized by lighting.

3. U–Pb SHRIMP analysis of zircon

Zircon grains were picked using a binocular microscope, and then mounted in epoxy resin for polishing prior to scanning electron microscope and cathodoluminescence imaging. The epoxy mounts were cleaned and gold-coated to give a uniform electrical conductivity for SHRIMP analysis. The zircon standard used was BR266 zircon (559 Ma, 903 ppm U). The isotopic composition of the U and Pb in zircons was determined using the SHRIMP II housed at Curtin University. Prior to spot analysis, rastering of the O^{-2} ion beam was carried out for 120–150 s to remove the gold coating and reduce the common Pb contaminant within the gold coating. A primary ion beam of 2.5–3 nA with a diameter of ~25 μm was focused onto the polished surface. The common Pb correction was carried out using the measured amount of ^{204}Pb. Isotope data were reduced using the SQUID2 software (Ludwig, 2009). Data were plotted on concordia diagrams using the Isoplot 3 software (Ludwig, 2008), with 2σ error ellipses on the concordia plots. All ages discussed in the text are reported at the 2σ confidence level. Data from SHRIMP analysis of zircon are presented in Supplementary Table A, together with paleomagnetic data. Zircon analyses that are discordant or marked by high common Pb contents ($>3\%$) are listed in Supplementary Table A, but were excluded from age calculations and further discussion.

A total of eight ash beds were sampled from the 23, 27, 151, 160, 223, 230, 292, and 483 m levels within the Ouberg Pass locality from the Waterford Formation (depths <98 m) and the Abrahamskraal Formation (depths >98 m). (Samples are named according to stratigraphic level, e.g. "OPA23" is Ouberg Pass ash bed from meter 23.) A compilation of all zircon age data indicate that the majority of grains (~60%) derive from Permo-Carboniferous to late Permian magmatic sources. All samples contain some xenocrysts, with a prominent secondary population ranging in age from 480 to 540 Ma (Fig. 8 top). Smaller populations of Neoproterozoic grains with no clear modal age are observed, together with single grains of Meso-Paleoproterozoic age. Ash beds differed greatly in the number of zircons recovered, with more abundant zircons found in samples with a smaller xenocrystic contribution. Few zircons were recovered from samples OPA23 and OPA27, and the age of the zircon grains suggests that the zircons are dominantly inherited from pre-existing sources.

Zircon grains from all samples are predominantly euhedral and idiomorphic with oscillatory cathodoluminescent zoning. Aspect ratios are typically in the 3–4:1 range, but many acicular grains with higher ratios are also observed, especially for the <300 Ma population. We do not observe rounded grains that would imply mechanical abrasion during transport, and metamorphic rims are absent, even for xenocrysts with ages >300 Ma. While not conclusive, these observations suggest that the presence of older, inherited material in the ash beds is due to magmatic processes, not post-eruption reworking.

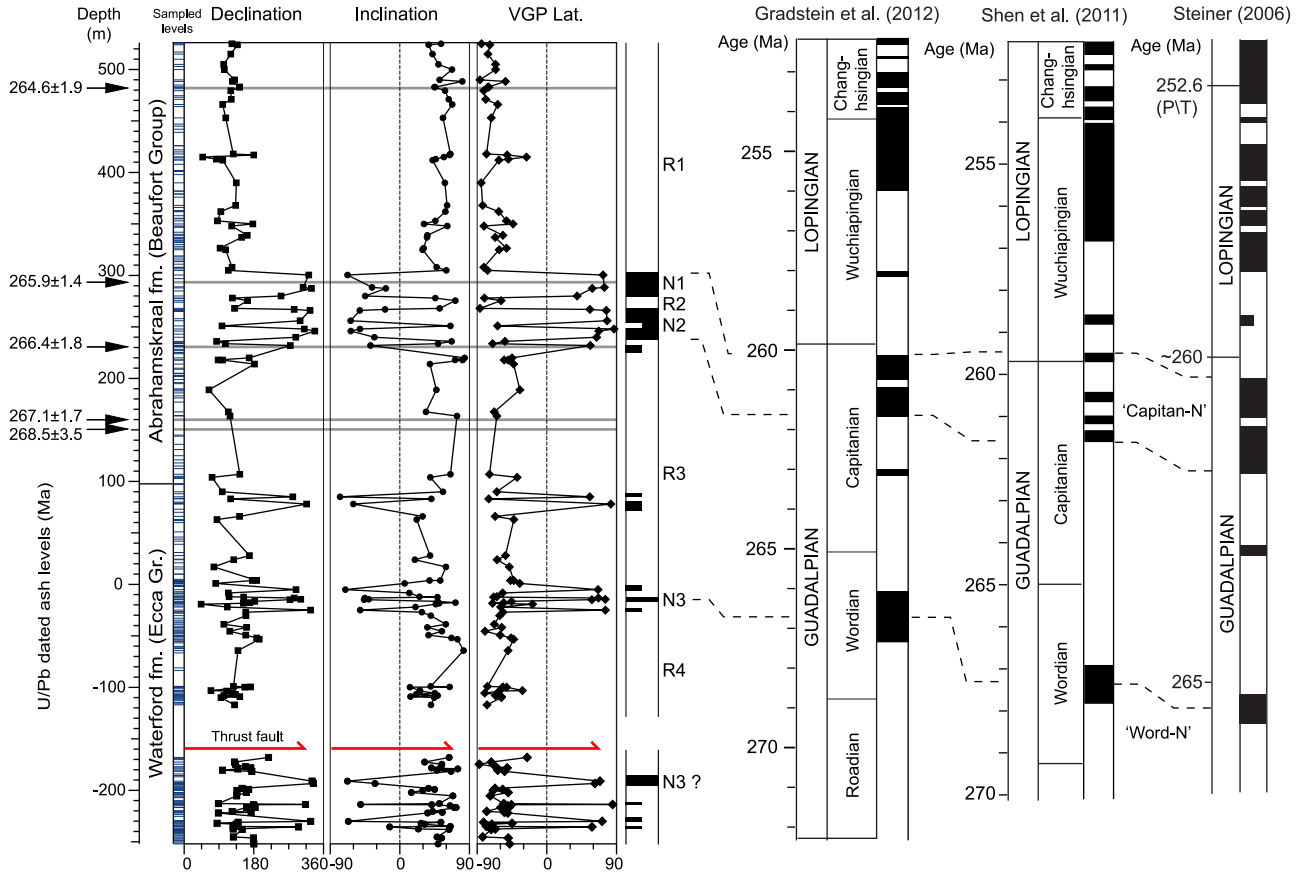


Fig. 7. ChRM directions, VGP latitude and magnetostratigraphic results from the Ouberg Pass section. Sampled levels are indicated and paleomagnetic directions of samples which gave good results are plotted. The polarity reversal sequence is compared with the composite from Steiner (2006), with the latest Permian time-scales from Shen et al. (2011) and Gradstein et al. (2012).

For samples with the most abundant zircon, the $^{238}U/^{206}Pb$ ages are dominantly Permian to Permo-Carboniferous. The age distribution appears homogeneous, with no discrete subpopulations (Fig. 8, bottom), suggestive of continuous, long-lived (ca. 20–30 Ma) magmatism in the source region. The modal peak of this youngest population is distributed around a mode of 267.5 Ma +2.1/−1.2, which may be regarded as the "average" age of eruption in the volcanic source region (Fig. 8). At the youngest end of the age spectrum, individual zircon grains are <260 Ma. Like the grains from the oldest end of the population, these youngest grains are generally outliers from the main population, We consider these grains to be of suspect age, given the potential for imperceptible Pb-loss from the standpoint of SHRIMP resolution, and a Pb-loss discordant trajectory that is parallel to the Concordia curve (e.g., Mundil et al., 2001, 2004). Likewise, we interpret zircon grains with ages >300 Ma as inherited, with no significance with respect to the age of eruption/sedimentation of the Abrahamskraal Fm. We will return to this point in the discussion section when comparing our interpreted age of the Beaufort Group with that of recent workers.

A total of five ash beds (OPA151, OPA160, OPA223, OPA230, and OPA483) yielded significant numbers of zircons with minor inheritance. Analytical results from these samples are shown in Fig. 9. The U–Pb results from the first four samples (OPA151, OPA223, and OPA230) yield "Concordia" ages, i.e., internally consistent, mean-weighted $^{207}Pb/^{206}Pb$–$^{238}U/^{206}Pb$ ages. The Concordia ages for the samples are as follows: OPA151, 268.4 ± 3.8 Ma (13/24 grains); OPA160, 263.3 ± 3.3 Ma (20/40 grains); OPA230, 267.9 ± 3.1 Ma (13/28 grains); and OPA-292, 266.4 ± 2.1, 16/23 grains. The reported ages for individual samples are very similar to the results obtained using the TuffZirc algorithm of Ludwig and Mundil (2002), demonstrating the reasonably objective criteria used to eliminate biased analyses from age calculation (i.e., inheritance, high analytical error, discordance, or possible, undetected Pb loss). The remaining sample OPA483 contains the highest proportion of inherited grains and consequently the least robust age, reflected in the higher MSWD and age uncertainty. For this sample, the reported age of 268.3 Ma $\pm$ 6 is the mean-weighted $^{238}U/^{206}Pb$ age from the population of the youngest seven grains. We interpret these ages from volcanic ashes as direct determinations of the age of sedimentary deposition for the Abrahamskraal Fm.

For the reported ages, there is no obvious younging trend that would fit with their known stratigraphic position. Geologically, the time required for deposition of the sequence between the dated levels could have taken place within roughly the age uncertainty of our SHRIMP-based age determinations (average $\sim$1.4% or 3–4 Ma). However, the known stratigraphic order of the samples (i.e. relative age) allows us to constrain the most probable age of the individual sample within the finite limits of analytical error associated with that sample. This problem is addressed by using a Monte Carlo simulation of best-fit ages for a specified range of stratigraphically ordered ages with finite errors (Ludwig, 2008). Applying this technique to the five samples obtains the following best fit ages: OPA 483, 264.6 Ma (+1.9/−4.3); OPA 292, 265.9 Ma (+1.4/−1.8); OPA230, 266.4 Ma (+1.8/−1.5); OPA160, 267.1 Ma (+1.7/−1.7); OPA151, 268.5 Ma (+3.8/−1.7). This implies a total age for the stratigraphic section of 3.9 Ma (+5.5/−2.6), corresponding to a depositional rate of $\sim$85 m/Ma.

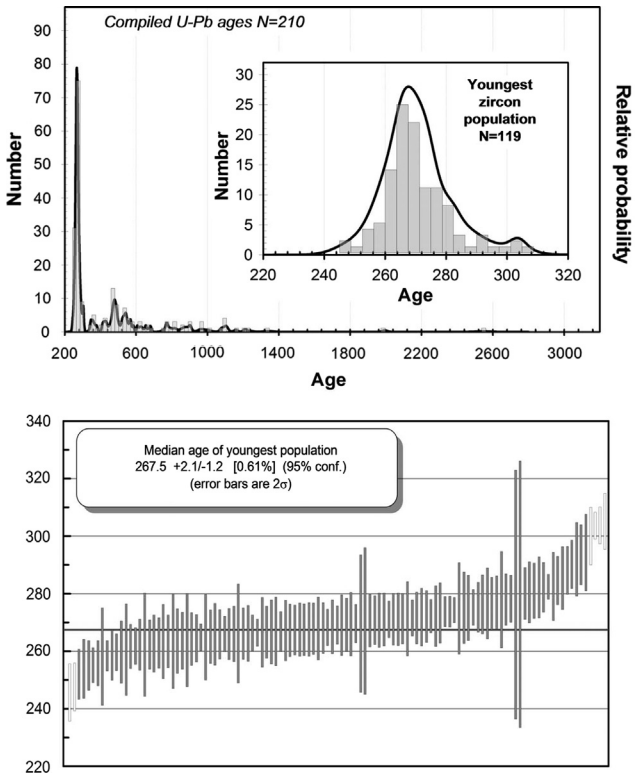


Fig. 8. Compiled U–Pb zircon dates from all studied tuff samples. Cumulative histogram indicates strong inheritance from both the late Mesoproterozoic Namaqua–Natal belt as well as contributions from 480 to 550 Ma aged belts, probably the Saldanian–Pampean or Famatinian orogens. More than 60% of the zircons are from <300 Ma, with a preponderance of grains in the 260–280 Ma age range suggestive of volcanic sampling of preexisting Permian volcanic rocks during eruption of the 264–268 Ma volcanoes.

4. Discussion

The rock-magnetism of the Ouberg Pass section demonstrates relatively minor variations in magnetic mineralogy, as evidenced by the S-ratio values close to 1. More noticeable changes are observed in the concentration of magnetic minerals, which is shown in the susceptibility and IRM logs. Values of S-ratio smaller than 1, indicating the presence of hematite, and hence suggesting a more oxidizing environment, occurs sporadically with frequency increasing upsection. Beds with high susceptibility and IRM intensity occur concomitantly above +160 m depth with frequency increasing upsection (Fig. 3). We suggest that this pattern represents the influence of a different sediment source, such as volcanic terrains, which are rich in iron-oxides and thus increase susceptibility and IRM. This change of physical properties of the rock can be easily detected in the field using a portable susceptibility meter and it is suggested that it could be used to map a conventional boundary between the Ecca and Beaufort Groups.

A magnetic polarity sequence constructed using the VGP latitude of individual samples shows a prevalent reversed polarity with 4 reversal and 3 normal magnetozones defined by at least 2 consecutive samples with the same polarity (Fig. 7). The presence of normal polarity zones indicates that the studied section does not belong to the Kiaman reversed superchron, thus it is attributed to the Illawarra mixed polarity zone compatible with the Late Permian age of the Beaufort Group. Although most workers place the top of the Kiaman reversal polarity interval at or above the base of the base of the Capitanian, the precise age is controversial (Gradstein et al., 2004; Menning et al., 2006; Steiner, 2006). Early compilations suggested an age of ~260–262 Ma for Illawaran normal zones (Opdyke and Channell, 1996), but more recent compilations by Gradstein et al. (2004) and Menning et al. (2006) cite an age of ca. 265 Ma based on a zircon age of 265.3 ± 0.2 Ma from ash beds below the Capitanian GSSP (Bowring et al., 1998). Unfortunately, the magnetostratigraphic study of these GSSP rocks was carried out separately and has been published in abstract form only (Menning, 2000).

Our new results from the Ouberg Pass support a correlation between our N3 normal zone and the Wordian ("Word-N") normal-polarity zone (Steiner, 2006; Shen et al., 2011; Gradstein et al., 2012) and correlate the subsequent N1 and N2 normal polarities with the P2, "Capitan-N" zone of Steiner (2006), equivalent to the normal polarity zones at the upper Capitanian in the geological time scale of Gradstein et al. (2012) and to the coeval prevalent normal polarity zone of Shen et al. (2011), as shown in Fig. 7. Beside the differences in absolute ages, the polarity sequence from Ouberg Pass can be correlated very well to both the composite magnetic polarity for late Permian of Steiner (2006) and the geological time scale of Gradstein et al. (2012) (Fig. 7). According to both these correlations the Ouberg Pass section represents a time span of about 7 Myr.

Combined geochronological and magnetostratigraphic results place the Ouberg Pass section in the Illawara mixed-polarity zone (Guadalupian). Extrapolating from the modeled age progression of ash beds (Fig. 9, bottom right) ages and taking magnetozone N3 as the base of Illawarra, we deduce an age of ~269 Ma for the end of the Kiaman reversed superchron, which appears to be ~4 Ma older than the ~265 Ma age from recent estimates. While the relative imprecision of SHRIMP ages is an impediment to more precise refinement of the age of the boundary for this study, we note that recent recognition of Pb-loss using an improved ID-TIMS technique suggests that the 265 Ma constraint is likely to be too young (Mundil et al., 2004; Shen et al., 2011).

Moreover, our results clash with the interpretation of Fildani et al. (2007, 2009) who obtained U/Pb radiogenic isotope ages between 275 Ma and 254 Ma in the Laingsburg Fm, middle Ecca Group, implying a much younger age (>10 Ma) for the Beaufort Group, which is stratigraphically higher (500–1000 m). Although Ouberg Pass region and the study area of Fildani et al. (2007, 2009) are located about 100 km apart, it seems unreasonable to attribute such a large age difference to diachroneity of sedimentation. The discrepancy between the stratigraphic ages we assign and those that would be inferred from the Fildani et al. (2007, 2009) is the result of different criteria for age interpretation. The ages that we report here represent mean-weighted data from a substantial population ($n>6$) of individual zircons, whereas Fildani et al. (2007, 2009) based their stratigraphic age assessment on the single, "youngest" grain in a given population of volcanic zircons. We contend that the mean-weighted age of a population is more indicative of the stratigraphic age since it is more statistically robust and automatically excludes older and younger outliers that are possibly inherited or affected by minor Pb-loss, respectively.

5. Conclusions

The rock-magnetism and magnetic fabric corroborated paleomagnetic measurements have provided new insights into the depositional history of the Beaufort Group. Rock magnetism indicates that the main carrier of the paleomagnetic remanence in the studied section is magnetite/maghemite. The presence of a significant amount of high-coercivity minerals such as hematite, which might account for the red/purple color of some sedimentary beds in the Abrahamskraal Fm., is generally not corroborated by the rock-magnetic analysis, which shows hematite is rarely a magnetically dominant phase even in red samples. A few hematite-bearing samples aside, the magnetic mineralogy does

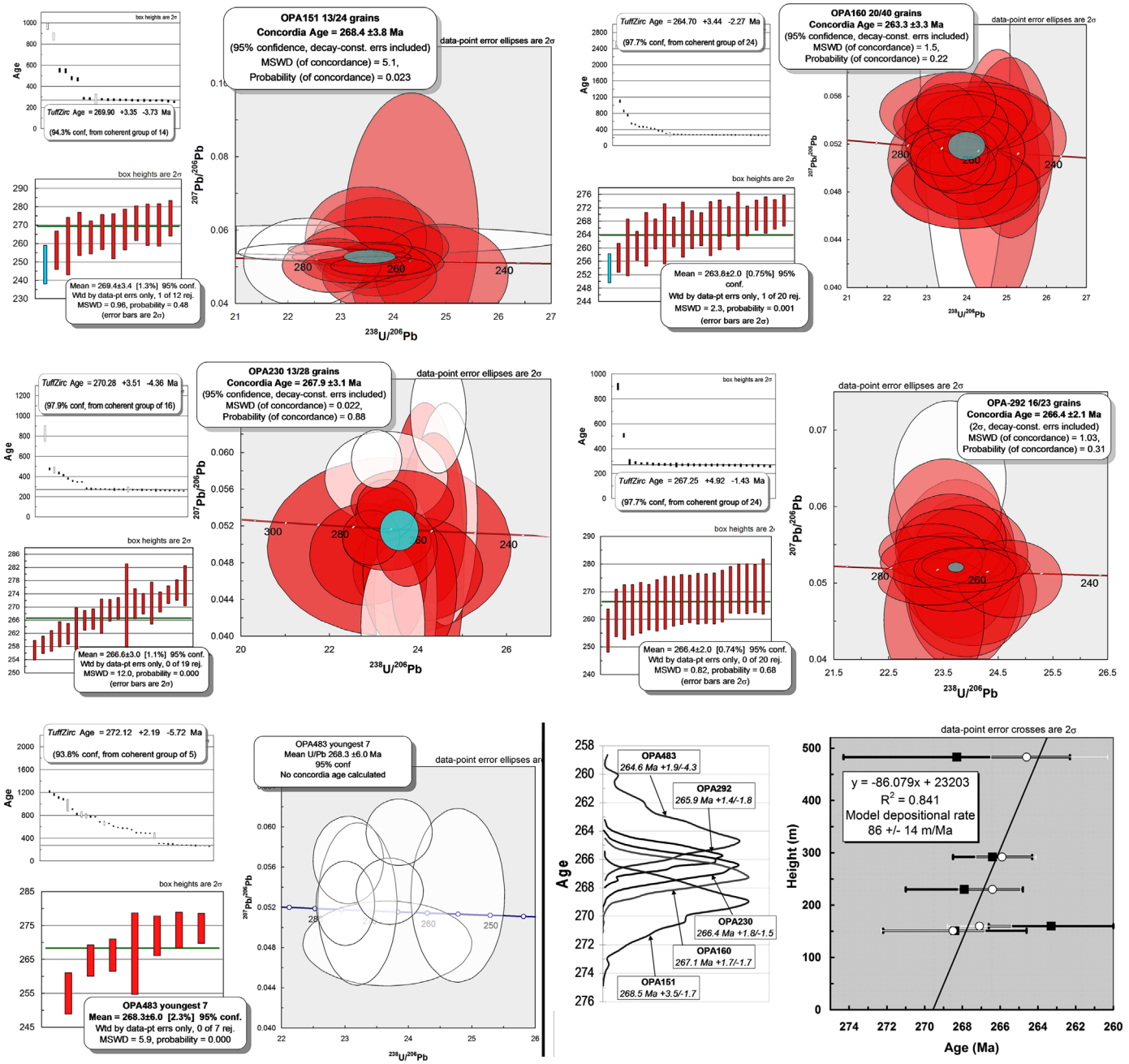


Fig. 9. Tera–Wasserburg type concordia plots of U–Pb data from individual samples from the Ouberg Pass tuff deposits. Bottom right figure depicts the depositional age model from Monte Carlo simulation of best fit ages of stratigraphically ordered samples. White dots indicate model age with error bars, and black squares and bold error bars depict the determined U–Pb age for individual ashbed samples.

not change dramatically through the lower part of the section. We do find a change in the concentration of magnetic minerals at the stratigraphic height of ~160 m corresponding to the first spike in magnetic susceptibility and IRM intensity (Fig. 3). This marks a change in the character of the magnetic properties of the sediments, which is distinctive of the upper part of the Abrahamskraal Fm., suggestive of a change in sediment provenance. This change could be explained by an enhanced influx of volcanic-rich terrigenous sediment into the fluvial sediments at this time. The change from relatively low variability to relatively high variability in magnetic susceptibility (Fig. 3) occurs above the contact from the Waterford Fm. to the Abrahamskraal Fm. and could be used as a mapping tool to define this contact in other parts of the succession.

This paleomagnetic and geochronological study from the continental Karoo succession contributes to an improvement of the Late Permian time scale. The NRM from the Abrahamskraal Fm. and uppermost Waterford Fm. shows a strong magnetic overprint (Component-A) whose direction does not correspond to the present-day field. Based on published age data (Jourdan et al., 2007) we hypothesize that such an overprint is related to the emplacement of the Karoo Large Igneous Province in the Western Cape region, which is well represented in the study area by widespread doleritic sills. This partial overprint of the NRM notwithstanding, the characteristic paleomagnetic directions (component-B) were preserved at relatively high unblocking temperatures ($T \geq 450$ °C) and could be isolated in a sufficiently large number of samples. The magnetostratigraphy of the Ouberg

Pass section (Fig. 7) shows a mixed magnetic polarity with the prevalence of reversed polarity zones interspersed with short normal magnetozones indicating a post-Kiaman age for these upper Permian rocks. U/Pb-based radiogenic isotope dating of zircon crystals extracted from 5 ash layers in the Ouberg Pass section, gave ages between 264 and 268 Ma. Despite some problems with inherited grains, the age of the mid-section can be conservatively assigned an age ca. 264–268 Ma. Based on the time frame outlined by the U/Pb zircon ages, the studied section is ascribed to the lower part of the Illawarra mixed polarity zone (Capitanian/Late-Guadalupian), showing a good correlation with the latest polarity-reversal scales (Steiner, 2006; Shen et al., 2011; Gradstein et al., 2012). Our geochronological results and paleomagnetic interpretation indicates an age for the "Capitan" normal interval (between 240 and 300 m) of about 269 Ma approximately 4–5 Myr older than previous estimates (Steiner, 2006; Shen et al., 2011; Gradstein et al., 2012).

The mean paleomagnetic direction of 308, −53 ($\alpha_{95}=4.1$) provides a precisely dated paleopole for southwestern Gondwana, obtained in the continental Karoo succession. Our paleopole position from the late Permian Ouberg section is indistinguishable from that reported by De Kock and Kirschvink (2004) for the P/T boundary. According to the elongation/inclination flattening correction of Tauxe and Kent (2004) the detrital remanence acquisition and/or sediment compaction has shallowed the mean inclination by 9°, as a consequence, the latitude of the shallowing-corrected paleopole remains practically the same but the longitude became appreciably different.

Acknowledgments

We would like to thank all persons involved in the Beaufort Project, in particular Anne Powell, A. Palfrey, T. Payenberg, J. Vermeulen, E. King, K. Ratcliffe, G. Hildred, D. Cole, A. Mistry, M. Yan, J, Hansma, and M. Danisik. The suggestions of two anonymous reviewers contributed to improve this paper. This work was supported by Chevron Australia Pty Ltd. and the Australian Research Council (LP0991834).

Appendix A. Supplementary information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.epsl.2013.05.017.

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